

3.3.3 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions, using the Quotient Rule when possible.

a) $y = \frac{x^2 + 4x + 3}{\sqrt{x}}$

b) $h(t) = \frac{1 + 2t}{3 - 4t}$

c) $g(u) = \frac{u + 1}{u^3 + u - 2}$

d) $y = \frac{t}{(t-1)^2}$

e) $w(r) = \frac{r^2 + \sqrt{r}}{2e^r + \pi}$

f) $A(s) = \frac{s^2 + 2}{s^4 - 3s^2 + \sin(1)}$

2. Find the equation of the tangent line to the curve at the given value of x .

a) $y = \frac{3x+1}{x^2+1}, x=1$

b) $y = \frac{\sqrt{x}}{x+1}, x = 4$

3. Find the equations of the tangent lines to the curve $y = \frac{x-1}{x+1}$ that are parallel to the line $x - 2y = 2$.

4. Determine the value of $h'(2)$, given that $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, & $g'(2) = 7$.

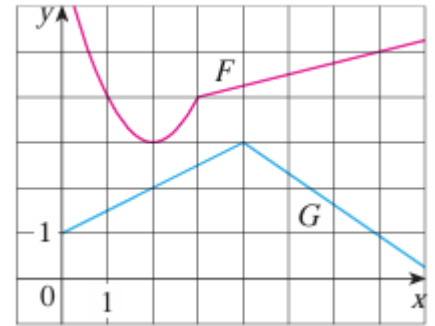
a) $h(x) = \frac{f(x)}{2g(x) + x^2}$

b) $h(x) = \frac{g(x) + 1}{2 + f(x)}$

5. If f is a differentiable function, find an expression for the derivative of the function $y = \frac{1 + xf(x)}{\sqrt{x}}$.

6. Let $P(x) = F(x)G(x)$ and $Q(x) = \frac{F(x)}{G(x)}$, where F and G are the functions whose graphs are shown below.

a) Find $P'(2)$.



b) Find $Q'(7)$.